Supplementary material for CON 3.1

Rohit Goswami[a,1]

[a] *TurtleTech ehf., Reykjavík, Iceland*

Abstract

These notes record the measurements behind the figures that accompany the CON 3.1 paper. The restart displacements, the format matrix, the frame bytes, the velocity step and the illegal-file classes are reads of the campaign files and the conformance corpus. The hash figure is a query of those bands. The ingest tables are the stored five-repeat measurements of that store.

S1. Restart difference

The interrupted band in Section **??** writes a checkpoint and then continues. Figure S1 compares each resumed movie with the uninterrupted movie at the same index. Column 3 is that checkpoint. The final column is the band after the five steepest-descent steps.

S2. Round-trip fidelity

Figure S2 uses two unmodified files. The matrix rows other than the velocity-file row are the first frame of the height-held EMT band. That frame already carries mask 4, a z-only hold, on the adatom, together with ids, forces and an energy. It carries no velocities and no time unit, so the time-unit cell on those rows is absent. The velocity-file row is a separate round trip of the unmodified 28-atom velocity fixture, whose time unit is the natural unit. Neither file was edited. The library conversion into ASE atoms passes energy and forces as positional arguments. ASE 3.29.0 accepts those values only as keywords, so the measurement supplies them as the keywords that constructor requires. The grades are the fields that come back from a write and a read after that conversion.

Email address: `rgoswami@ieee.org` (Rohit Goswami)
[1]ORCID: 0000-0002-2393-8056.

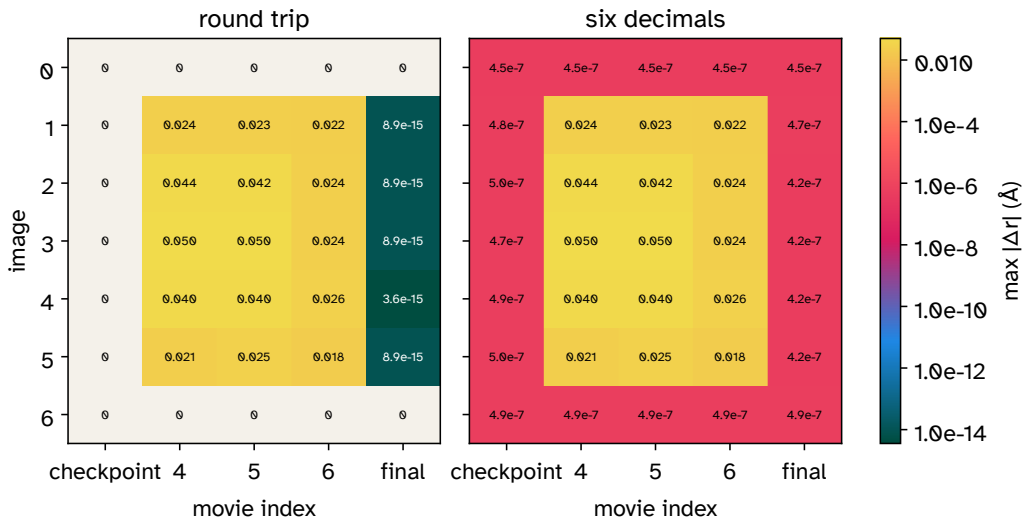


Figure S1: Minimum-image maximum component of the displacement, in Å, between the uninterrupted Morse band and each resume. The checkpoint column is bitwise exact for the round-trip printer and about 5×10^{-7} Å for six decimals. The final column is at most 8.9×10^{-15} Å for the round trip and 4.9×10^{-7} Å for six decimals. Intermediate columns differ by up to 0.050 Å because the search has taken further steps after the load.

S3. Frame bytes and parse time

The frame is the 28-atom velocity file, 3,826 bytes. The timed files are 400 copies of that frame and 400 copies with the velocity section removed. The reported times are medians of 31 paired reads.

S4. Velocity mislabel

The kinetic energies and the Verlet step are the illustration in Section ??. The conversion from the natural unit to femtoseconds is the factor used for that illustration. The cell is the slab cell of the file.

S5. Illegal-file classes

The eight illegal files are the conformance corpus named in Section ??. readcon raises a message that names the class. ASE 3.29.0 reads the same files with its eon reader.

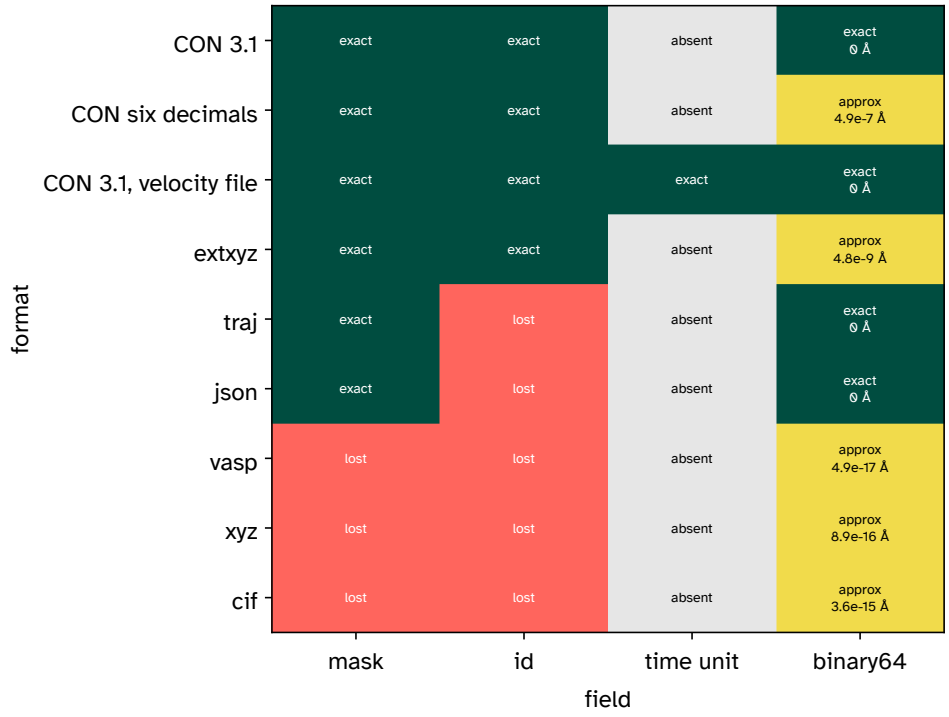


Figure S2: Fields kept by a write and a read. The held-frame rows have no time unit in the source file. The velocity-file row keeps the natural unit. CON 3.1 keeps the mask, the id and binary64 on both files. Six decimals keep the mask and the id and move a held-frame coordinate by 4.9×10^{-7} Å. Extended XYZ keeps the mask and the id and moves a coordinate by 4.8×10^{-9} Å. The trajectory and JSON writers keep the mask and the coordinates and drop the id. POSCAR, XYZ and CIF drop the mask and the id. The number in the binary64 cell is the maximum absolute coordinate change. A grey cell is a time unit the source frame did not carry.

S6. Database hashes of the campaign bands

The hashes are a query against readcon-db 0.1.8. The five bands were appended as stored trajectories, and each frame hash was read back by the same query. The bands are the uninterrupted Morse restart, its round-trip resume, its six-decimal resume, and the two EMT mask bands of Section **??**. The query returns 35 images and 35 hashes. Every returned hash locates the frame that was stored. Endpoint coordinates of the uninterrupted band and the round-trip resume agree: the maximum absolute component is 0 at both endpoints and 8.9×10^{-15} Å over the seven images. The stored texts

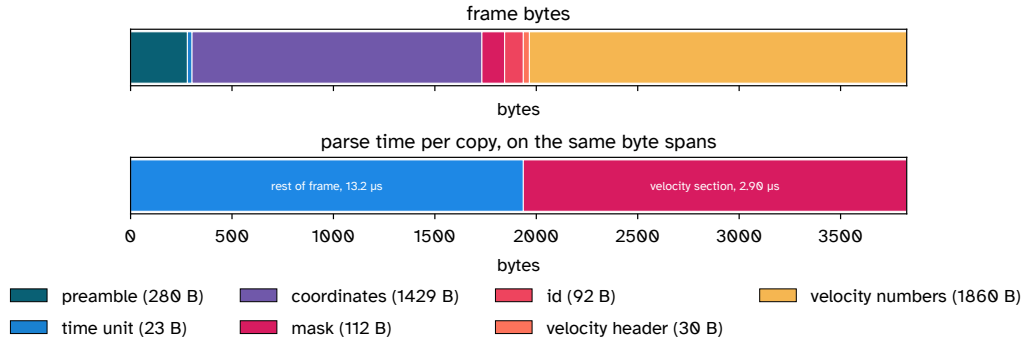


Figure S3: Byte widths of one CON 3.1 frame, and the parse time of the same spans. Masks, ids and the time unit are 227 bytes, against 1,429 coordinate bytes and 1,860 velocity-number bytes. The velocity section adds 2.90 μs to a 13.2 μs read of the rest of the frame. Masks and ids share the coordinate line, so their parse time is inside that 13.2 μs.

still differ, so the hash keeps every frame apart. On the reactant the only differing line is the metadata line. The printed curvature products differ by one byte, 2,507 bytes against 2,506, and the coordinate rows match. The product frame differs only in that same metadata line. The five intermediate images differ in the metadata line and in coordinate rows at the last printed digits, which is the 8.9×10^{-15} Å displacement. Figure S7 is the same five bands on the (s, d) plane [1].

S7. Failure cases: frame, grading and drift files

The survey frame is a Pt(100) $2 \times 2 \times 2$ slab with 6 Å of vacuum and one Pt adatom 1.8 Å above the hollow site. Every coordinate carries a uniform offset of up to 10^{-3} Å from a fixed seed, so all seventeen significant digits are needed. The bottom four atoms are fixed on every axis, and the adatom is fixed in z. The ids are 17, 3, 11, 5, 23, 2, 13, 7 and 41 in row order. Velocities are normal deviates of standard deviation 4×10^{-3} in the natural unit, zero on held axes. For ASE the masks become one FixAtoms and one FixCartesian, the ids an integer array named `id`, and the velocities ASE momenta. For MDAnalysis the velocities are converted to Å/ps, its base unit. We count a field as kept when it reads back equal; a mask counts as a full freeze when every masked atom returns frozen on every axis. A velocity is declared when the file names its unit and the values agree after conversion to within 10^{-6}

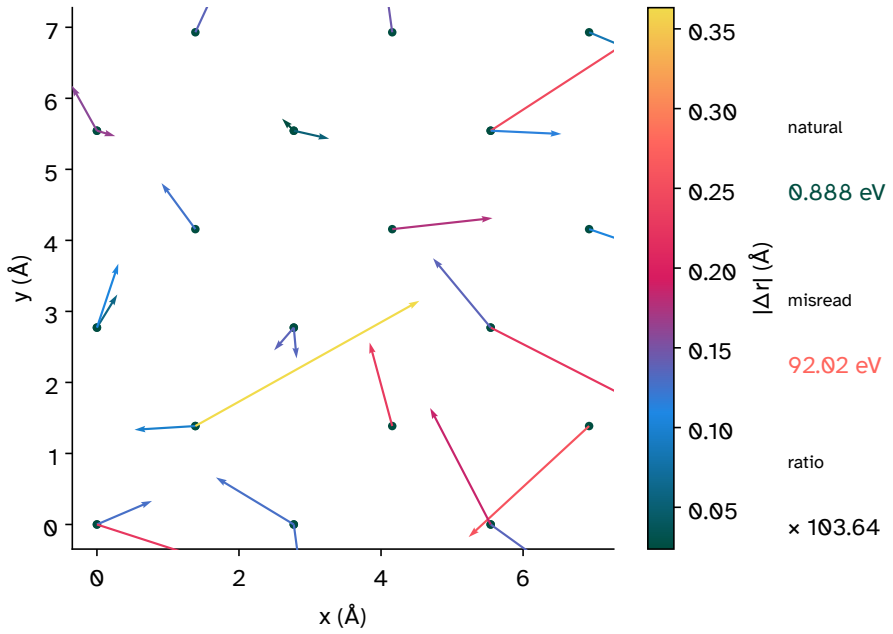


Figure S4: Displacement after one velocity Verlet step of 1 fs on the 28-atom slab when the natural velocity is misread as Å/fs. The kinetic energy goes from 0.888 eV to 92.02 eV, a factor of 103.64. The RMSD is 0.096 Å and the maximum component is 0.315 Å. The arrows are drawn at 10 times that displacement.

relative. When the file names no unit, we report what a reader that takes the numbers as Å/fs would see, a kinetic energy 103.64 times too large.

Each drift file has three Pt atoms with the header

```
Lattice="5.5437171645025325 0.0 0.0 0.0 5.5437171645025325 0.0 0.0 0.0
    20.0" Properties=species:S:1:pos:R:3:move_mask:L:3:id:I:1 pbc="T T F"
```

and rows `Pt 0.0 0.0 6.0 T T T 17`, `Pt 2.7718585822512662 0.0 6.0 T T T 3` and `Pt 1.3859292911256331 1.3859292911256331 7.8 F F T 41`. The variants drop the `pbc` key, replace the header by a plain comment and the rows by their first four fields, write the z value 6.0 as `6.0d0`, spell the booleans `True` and `False`, add `note="E=1.5 eV"`, and add `units="angstrom,eV,fs"`. The readers are ASE 3.29.0, the libAtoms extended XYZ parser 0.4.6 through its C path and its Python path, chemfiles 0.10.4, and MDAnalysis 2.10.0. The correctly rounded binary64 value of 2.7718585822512662 has the bit pattern `40062cc4313f3fb9`; chemfiles returns `40062cc4313f3fb8`.

The resume study uses an EMT Pt(100) $3 \times 3 \times 3$ slab with 6 Å of vacuum,

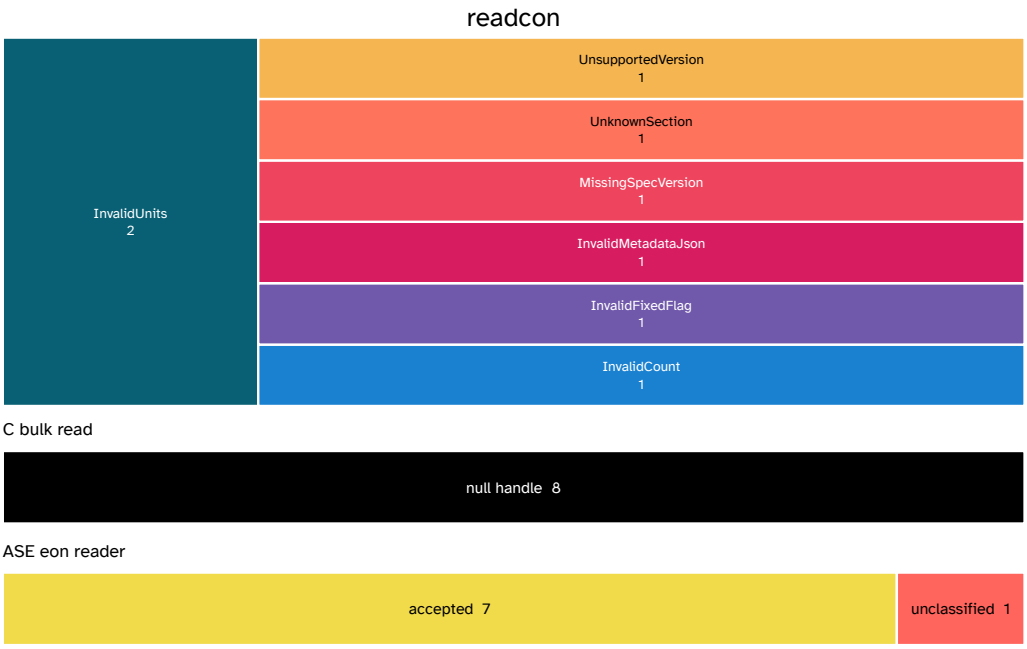


Figure S5: Classes on the eight illegal files. readcon names a class for each file. The ASE eon reader accepts seven of the files and stops without a class on the per-type count of zero.

the bottom layer held, and one adatom 1.75 Å above a hollow and held in z, hopping to the neighbouring hollow. Endpoints were relaxed with BFGS to 0.002 eV/Å. Five movable images were interpolated with the constraints applied. The band uses the improved tangent with a climbing image, and each step adds 0.02 Å^2/eV times the projected force. We interrupt after step 25. Each route writes all seven images and reads them back with the reader the route names, keeping whatever constraints and periodicity that reader returns.

S8. Reader rules

Table S1 collects the edge cases a reader applies. They are the classes of Section 3, stated as input and outcome.

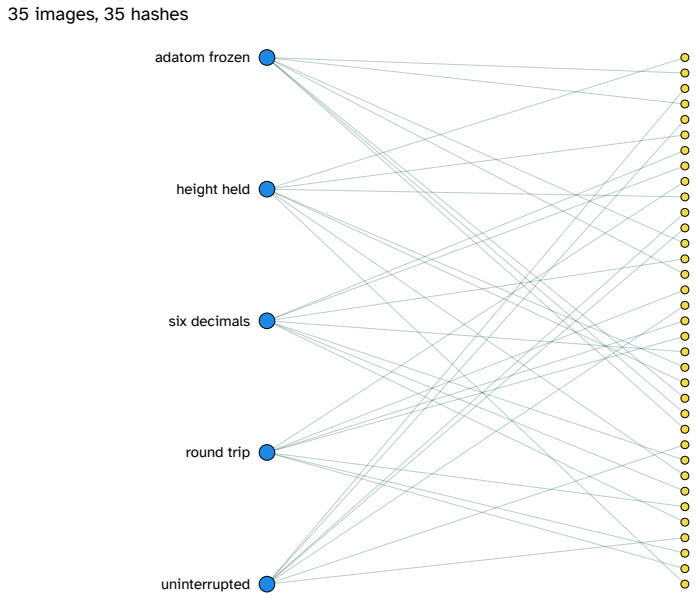


Figure S6: Campaign bands on the left and the database hash of each stored frame on the right. The query returns 35 images and 35 hashes. No hash is shared. The endpoints match in coordinates. Their hashes differ because the metadata line does.

S9. Paired peer timings on every payload

Table S2 extends the peer table of the main text to the 1-frame and 100-frame files. The comparison calls the Python binding with one worker, and ASE, chemfiles, MDAnalysis and h5py run in the same process with OpenMP and BLAS limited to one thread. XYZ and H5MD peers receive the coordinates of the same frames, written before the timed loop starts. Each cell is one-worker readcon-core, then the peer, thirty times, after one untimed call of each side. A repeat over 240 s would be recorded as a timeout and a peer that raises as a failure; neither occurred. The interval is a bias-corrected and accelerated bootstrap of the median with 9999 draws [2]. On one frame the fixed cost of opening the file dominates, which is why h5py and chemfiles lose their lead there.

S10. Stored ingest timings

readcon-db answers composition, energy and maximum-force queries over CON frames that an adaptive kinetic Monte Carlo campaign has already

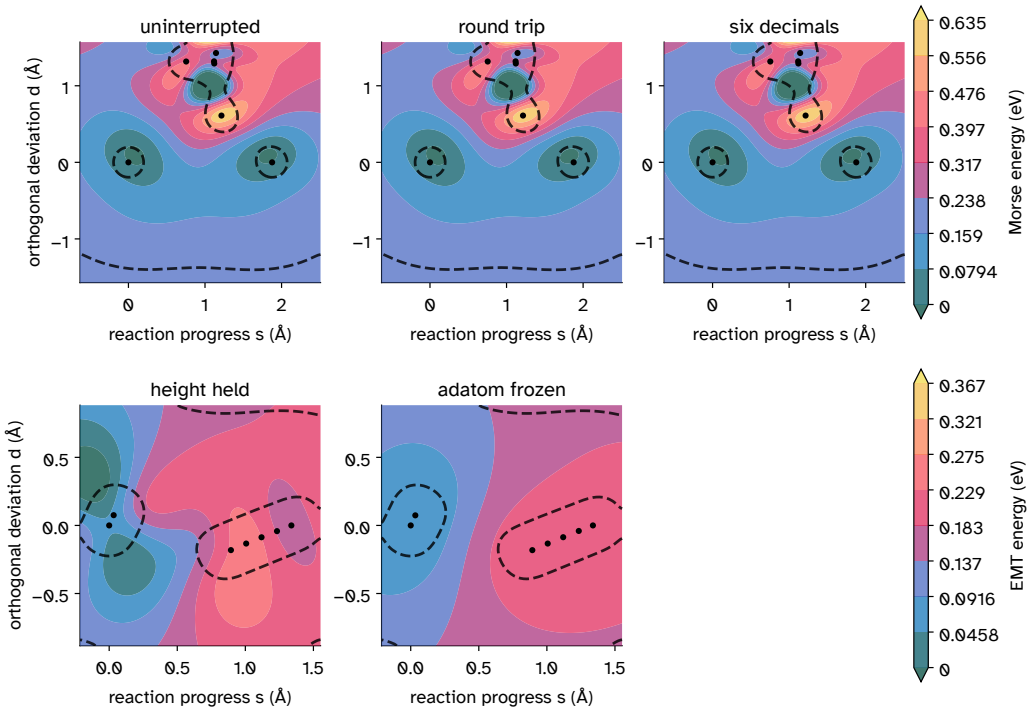


Figure S7: The same five bands on the (s, d) plane [1]. The Morse maps share one colour scale and the EMT maps share another. Black dots are the images. The round-trip and six-decimal maps sit on the uninterrupted Morse band at this scale.

written. The frame defined in the main text is unchanged. The numbers in this section are the stored five-repeat measurements. They were not remeasured for this manuscript. The machine for those measurements was an Intel Xeon with 8 CPUs and 47.07 GiB of memory. The software was Python 3.12.3, rustc 1.88.0, readcon-db 0.1.8, readcon-core 0.16.2, ASE 3.29.0 and numpy 2.5.4. Threads were pinned to one. Five repeats were interleaved, and each value is the median, followed by the minimum and the maximum. A megabyte is 10^6 bytes. A cold query follows a hint that cached pages are not needed, and then runs in a fresh process. A warm query is the next query in that same process. The timed corpora repeat ASE effective-medium searches: a hydrogen cyanide band, a copper adatom band, a platinum-hydrogen band and a platinum adatom dimer. Every frame in those corpora has its own hash, at both 1,000 and 10,000 frames. The clock starts before the write and stops after a composition check. Every timed query was kept only after the

Table S1: Reader rules for flags, versions, identity and incomplete input.

Input	Rule
Column 4 outside the legal set	Illegal flag.
Version 1 column 4 equal to 1	Fixed on x, y and z.
Version 1 column 4 other than 0 or 1	Illegal flag.
Missing id column	Id is the zero-based row index.
Repeated id	File order is kept.
Unique ids	Ascending id is the pre-grouping order.
Unknown major	Rejected.
Unknown minor of a known major	Accepted.
Unknown metadata key	Kept on rewrite.
Unknown section at this minor	Rejected.
Unknown section at a newer minor, width declared	May be skipped.
Non-finite number	Rejected.
Type count of zero	Legal empty frame.
Short header or short section	No partial frame is returned.
Empty first header line	User line, kept.
Empty second header line	Implicit version 1.
Blank line after a frame, no velocity block	Line 1 of the next frame.
Several gzip members	Read in order.

hit count and the sum of frame indexes matched an oracle.

Table S3 is the ingest of 10,000 frames. Inside a single ASE transaction the writer still commits every 5,000 rows, so 1,000 frames is one commit and 10,000 frames is two commits, plus the commit that closes the block. The other ASE column commits on every write. At 1,000 frames the medians are 0.0329 s (30,400 frames/s) for readcon-db, 0.646 s (1,550 frames/s) for one transaction, and 5.03 s (199 frames/s) for a commit on every write. At 10,000 frames readcon-db writes at 35.1 MB/s (33.0 to 38.0), the single transaction at 1.83 MB/s, and the per-write commits at 0.226 MB/s.

On composition, energy and maximum force at 10,000 frames, readcon-db takes about 8 to 10 ms when cold and about 0.2 ms when warm. The cold medians are 8.14 ms (5.12 to 10.7), 10.1 ms (8.11 to 14.4) and 9.57 ms (7.70 to 11.6), in that query order. The warm medians are 0.176 ms, 0.211

Table S2: Paired time ratios. The ratio is the readcon-core time divided by the peer time, so a value below 1 means readcon-core finished sooner. The interval is a bias-corrected and accelerated 95 percent interval of the median. Failures stay in the table and are not converted into times.

Payload	Peer	OK	Fail	Ratio	Interval
1 frame	ASE eon	30	0	0.153	[0.15, 0.155]
1 frame	chemfiles XYZ	30	0	0.893	[0.888, 0.902]
1 frame	h5py	30	0	0.365	[0.353, 0.37]
1 frame	MDAnalysis H5MD	30	0	0.0207	[0.02, 0.0223]
100 frames	ASE eon	30	0	0.235	[0.226, 0.238]
100 frames	chemfiles XYZ	30	0	1.29	[1.26, 1.32]
100 frames	h5py	30	0	14.1	[13.6, 14.8]
100 frames	MDAnalysis H5MD	30	0	0.134	[0.127, 0.139]
10000 frames	ASE eon	30	0	0.337	[0.335, 0.345]
10000 frames	chemfiles XYZ	30	0	2.15	[2.14, 2.17]
10000 frames	h5py	30	0	67.1	[66.5, 67.3]
10000 frames	MDAnalysis H5MD	30	0	0.207	[0.204, 0.213]
Pt slab	ASE eon	30	0	0.233	[0.231, 0.241]
Pt slab	chemfiles XYZ	30	0	1.14	[1.13, 1.16]
Pt slab	h5py	30	0	6.05	[5.91, 6.27]
Pt slab	MDAnalysis H5MD	30	0	0.394	[0.387, 0.404]

ms and 0.208 ms. The ASE database takes 0.286 s, 0.223 s and 0.225 s on the cold queries, and 0.258 s, 0.203 s and 0.206 s when warm. A directory of the same CON text, in 64-frame chunks, takes 0.193 s, 0.193 s and 0.197 s cold, and 0.140 s, 0.138 s and 0.138 s warm. Figure S10 shows both corpus sizes.

readcon-db occupies 2.50 times the raw CON text, at both 1,000 and 10,000 frames. The ASE database is smaller than the raw text. Figure S11 also shows gzip and zstd of the same text.

One adaptive kinetic Monte Carlo trace of 24,000 frames stores 27.9 MB of blobs, 27,864,917 bytes, equal to the raw file, because the stored texts are kept separately. The hash index of that stored trace records 7 unique structures, and their text is 8319 bytes. Those seven texts are the unhopped hydrogen cyanide state and its saddle, the unhopped copper state and its saddle, and both platinum-hydrogen states together with that saddle. Figure S12 compares the full trace with that unique set. That collapse is a property of the repeated states in the trace. It is separate from the campaign-band query in Section S6, which found no shared hash.

Table S3: Ingest of 10,000 frames from the stored measurements. Each cell is the median, then the minimum and the maximum of five interleaved repeats.

Writer	Time (s)	frames/s
readcon-db	0.311 (0.287 to 0.331)	32,100 (30,200 to 34,800)
ASE, one transaction	5.99 (5.88 to 6.29)	1,670 (1,590 to 1,700)
ASE, commit per write	48.4 (45.5 to 50.7)	207 (197 to 220)

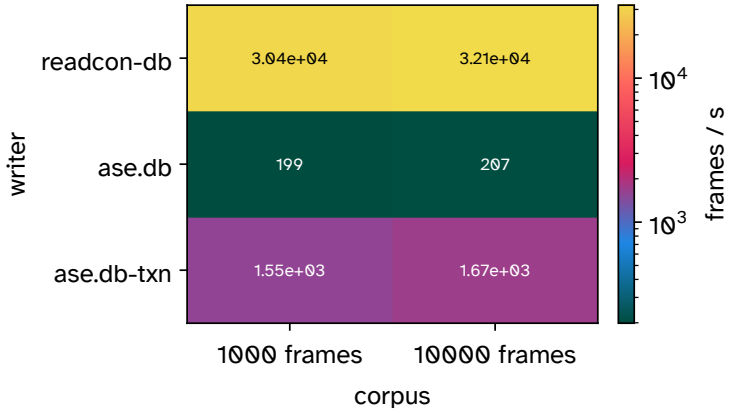


Figure S8: Median ingest rate from the stored measurements. Rows are readcon-db, an ASE database that commits on every write, and one ASE transaction. Columns are the two corpus sizes.

Nine correctness checks pass on that store. They cover a round trip, identical bytes sharing one hash entry, a one-digit coordinate change, a missing force or velocity section, periodicity taken only from the declared mask, a rebuild of the indexes from the stored text, a second rebuild, an ignored numeric cache, and a hash entry that remains after a duplicate is rewritten. The map is capped at 2 GiB, and one million frames does not fit.

S11. ABNF of one frame

The non-normative ABNF of one frame follows. An empty first header line is the user line, an empty second header line is implicit version 1, and a blank line after a frame is the next frame unless it begins a velocity block. The listing has 50 rules. The specification wins where the two disagree.

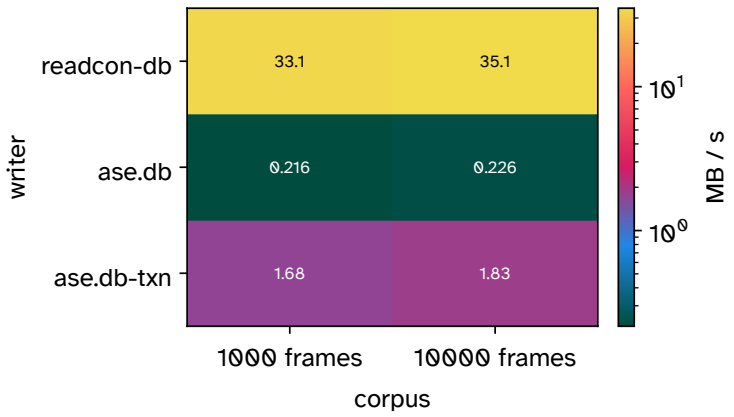


Figure S9: Median ingest bandwidth for the same three writers and the same five repeats. A megabyte is 10^6 bytes.

Listing S1: Non-normative ABNF for one frame.

```
file = [bom] 1*frame *(*ws nl)
bom = %xFEFF
; An empty free_line on line 1 is the user line.
header_open = free_line nl metadata_line nl float3_line float3_line
    free_line nl free_line nl
frame = header_open (zero_atom / typed_body)
free_line = *(%x00-09 / %x0B-0C / %x0E-FF)
metadata_line = *ws (json_object / legacy_metadata)
; An empty legacy_metadata line is implicit version 1.
legacy_metadata = *ws [(%x00-09 / %x0B-0C / %x0E-7A / %x7C-FF) *(%x00-09 /
     %x0B-0C / %x0E-FF)]
json_object = "{" *ws [json_pair *(*ws "," *ws json_pair)] *ws "}" *ws
json_pair = json_string *ws ":" *ws json_value
json_value = json_string / json_number / json_object / json_array / "true"
     / "false" / "null"
json_array = "[" *ws [json_value *(*ws "," *ws json_value)] *ws "]"
json_string = DQUOTE *json_char DQUOTE
json_char = "\" DQUOTE / "\\" / "\/" / "\b" / "\f" / "\n" / "\r" / "\t" /
    ("\u" 4HEXDIG) / (%x20-21 / %x23-5B / %x5D-FF)
json_number = ["-"] 1*DIGIT ["." 1*DIGIT] [("e" / "E") ["+" / "-"] 1*DIGIT
    ]
float3_line = *ws float 1*ws float 1*ws float *ws nl
count_line = *ws positive_integer *ws nl
count_list_line = *ws positive_integer *(1*ws positive_integer) *ws nl
float_list_line = *ws float *(1*ws float) *ws nl
zero_integer = 1*"0"
```

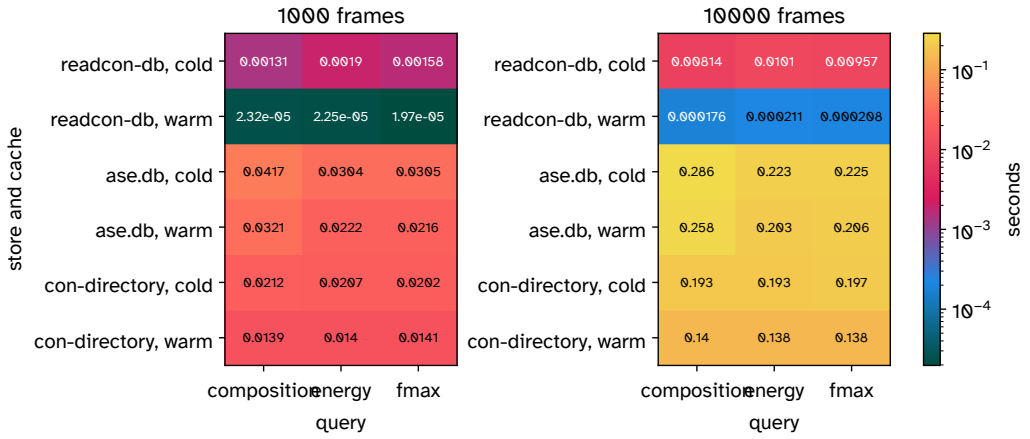


Figure S10: Median query time for composition, energy and maximum force, from the stored measurements. Each panel is one corpus size. Rows pair the store with a cold or warm cache.

```
zero_count_line = *ws zero_integer *ws [nl]
blank_line = *ws nl
zero_atom = zero_count_line (blank_line (blank_line / "") / "")
typed_body = count_line count_list_line float_list_line 1*type_block *
    section
type_block = symbol_line nl coord_label nl vector_atom_line *(nl
    vector_atom_line) [nl]
symbol_line = *ws symbol *ws
symbol = ALPHA *(ALPHA / DIGIT / "_")
coord_label = *ws "Coordinates of Component" 1*ws positive_integer *ws
vector_atom_line = *ws float 1*ws float 1*ws float 1*ws integer [1*ws
    integer] *ws
scalar_atom_line = *ws float 1*ws integer 1*ws integer *ws
atom_line = vector_atom_line
; A blank_line is a section only when a block follows, including a
    velocity block.
; Otherwise that line is line 1 of the next frame.
section = blank_line (1*section_type_block_vector / 1*
    section_type_block_scalar / 1*generic_type_block)
section_type_block_vector = symbol_line nl section_label_vector nl
    vector_section_line *(nl vector_section_line) [nl]
section_type_block_scalar = symbol_line nl section_label_scalar nl
    scalar_atom_line *(nl scalar_atom_line) [nl]
generic_type_block = symbol_line nl generic_label nl token_line *(nl
    token_line) [nl]
generic_label = *ws symbol *(1*ws (symbol / positive_integer)) *ws
```

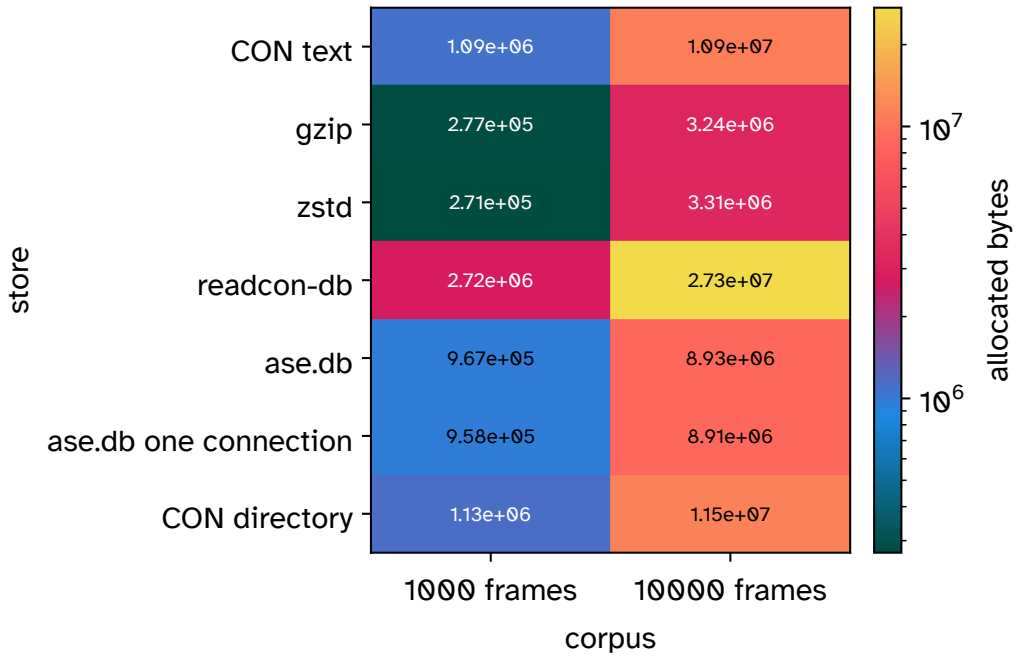


Figure S11: Allocated bytes for the raw text, two compressed copies, readcon-db, both ASE database files and a directory of 64-frame chunks. The readcon-db cell is the memory-mapped file, text plus indexes.

```
token_line = *ws (float / integer) *(1*ws (float / integer)) *ws
vector_section_line = *ws float 1*ws float 1*ws float 1*ws integer 1*ws
    integer *ws
section_kind_vector = "Velocities" / "Forces" / "Magmoms" / "Displacements
    " / "Spreads"
section_kind_scalar = "Energies" / "Charges" / "Spins"
section_kind = section_kind_vector / section_kind_scalar
section_label_vector = *ws section_kind_vector 1*ws "of Component" 1*ws
    positive_integer *ws
section_label_scalar = *ws section_kind_scalar 1*ws "of Component" 1*ws
    positive_integer *ws
float = finite_float / non_finite
finite_float = ["+" / "-"] (1*DIGIT ["." *DIGIT] / "." 1*DIGIT) [("e" / "E
    ") ["+" / "-"] 1*DIGIT]
non_finite = ["+" / "-"] ("infinity" / "nan" / "inf")
integer = 1*DIGIT
positive_integer = (%x31-39) *DIGIT
ws = SP / HTAB
NL = CRLF / LF
```

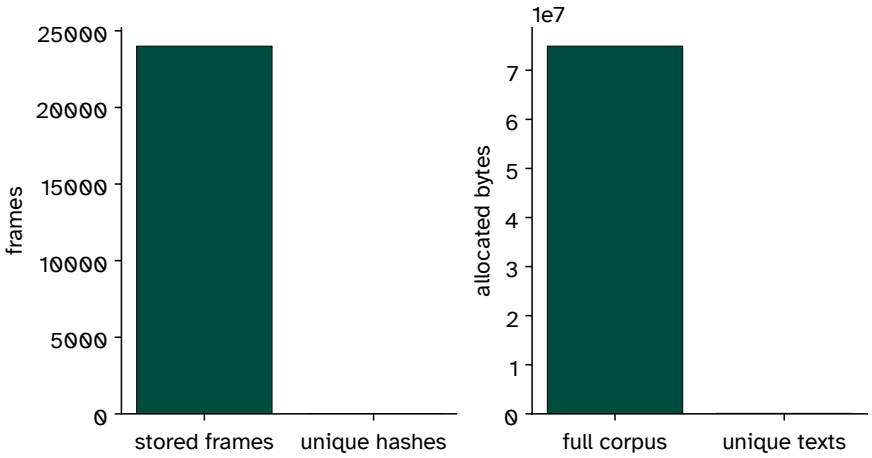


Figure S12: The stored 24,000-frame trace against its 7 unique structures. The left panel counts frames. The right panel is allocated bytes for the full corpus and for a corpus of the unique texts.

```
nl = NL
```

References

[1] R. Goswami, Two-dimensional RMSD projections for reaction path visualization and validation, MethodsX 16 (2026) 103851. doi:10.1016/j.mex.2026.103851.
URL https://linkinghub.elsevier.com/retrieve/pii/S2215016126000683

[2] B. Efron, Better bootstrap confidence intervals, Journal of the American Statistical Association 82 (397) (1987) 171–185. doi:10.1080/01621459.1987.10478410.
URL http://www.tandfonline.com/doi/abs/10.1080/01621459.1987.10478410